COMMERCIAL ITEM DESCRIPTION

Test Set, Insulation Breakdown AN/GSM-6( )

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for MIL-T-55687(EL) which is canceled.

This Commercial Item Description (CID) delineates the Salient Characteristics (Section A) and the Quality Assurance Requirements (Section B) of commercially available Insulation Breakdown Test Set, modified, as necessary, to meet the requirements specified herein, that is capable of providing facilities for measuring DC current at voltages up to 60 Kilovolts and provides a high DC breakdown test voltage for testing cables, bushings, transformers and generators for dielectric adequacy

A SALIENT CHARACTERISTICS

1. Safety Unless otherwise specified herein, the equipment shall comply with the Type II safety requirement specified in MIL-T-28800D

1.1 Personnel Hazards Personnel hazards shall be kept to a minimum The criteria of MIL-T-28800D, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract Verification that compliance with this requirement has been achieved shall be through a visual inspection (see para 1.6 below)

1.2 Edge Rounding Exposed Exposed edges and corners shall be rounded sufficiently to minimize laceration/puncture hazards, the design goal being a minimum radious of 0.04 inch (1 mm) for edges and a minimum of 0.5 inch (12 7 mm) for corners

1.3 Radioactive Materials Radioactive materials shall not be used (e.g , luminous dials/markings, electron tubes, surge arrestors and lenses)

DISTRIBUTION STATEMENT A. Approved for public release; distribution unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

1 4 Weight Caution Label All equipments over 35 pounds shall have a weight caution label for Two-Person Lift centered at the top surface The format and color code shall be in accordance with American National Standards Institute (ANSI) Standard Z35 11972 Label size shall be appropriate for high visibility and legibility Marking requirements shall conform to MIL-M-13231 The Weight Caution Label should read, for example

CAUTION
* Pounds
TWO-PERSON LIFT

* Weight of Test Set

1 5 Safety Features The following additional safety features shall be built into the equipment

a Indicating light for AC Power on.
b Indicating light for high voltage on
c Zero start interlock
c Protection against damage to any component of the equipment from short circuits at the output
e Provisions for external interlock and safety switches

1 6 System Safety Engineering Inspection A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 1, 1 1, 1 2, 1 3, 1 4 and 1 5 have been incorporated in the equipment Inability to meet these requirements shall constitute a failure of the test Results of this inspection shall be included in the Safety Assessment Report (DI-SAFT-80102) if required in the contract

2 Design and Construction Unless otherwise specified herein, the equipment shall comply with the Type II, Class 5, Style E, Color E, design and construction requirements of MIL-T-28800D All requirements which are automatically invoked by MIL-T-28800D to all types, and specifically to Type II, apply

2.1 Mainframe Plug-in Concept Equipments utilizing externally accessible, removable subassemblies (mainframe plug-in concept) does not meet the requirements of this CID.

2.2 Controls All controls required to operate the equipment throughout its specified performance range shall be located on the face of the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

3. Parts, Materials, and Processes Unless otherwise specified herein, the equipment shall comply with Type II parts, materials, and processes requirements of MIL-T-28800D.

3.1 Restricted Material. Equipment shall comply with the restricted material requirements of MIL-T-28800D including the requirement for radioactive deterioration of any material contained in the equipment due to effects of outgassing

3.2 Interchangeability Unless otherwise specified herein, the equipment shall comply with Type II interchangeability requirements of MIL-T-28800D.

3.2.1 Marking All replaceable assemblies, subassemblies, modules and printed circuit card assemblies shall be indelibly marked with the Federal Supply Code for Manufacturers (FSCM) and the contractor's part number

4. Marking and Identification Unless otherwise specified herein, the equipment shall comply with Type II marking and identification requirements of MIL-T-28800D

4.1 Supplemental Identification Plate The supplemental identification plate specified in MIL-T-28800D shall contain the following data:

a Nomenclature
b Contract Number
c Serial Number
d National Stock Number
e. US

4.2 Information Plate An information plate affixed to each deliverable equipment shall contain the following data

a Warranty termination date.
b. Brief statement of warranty limitations (for example opening a cover voids warranty, calibration adjustments may be made without voiding warranty, defective modules may be returned for warranty service, etc)

4.3 Battery Warning Label Equipment containing batteries shall provide a battery warning indication in accordance with MIL-T-28800D, para 3.10.3.1.3. This label shall also inform operators of hazards associated with batteries (e.g., do not charge batteries at certain temperatures). The warning label shall also list health hazards due to chemicals in the battery.

5. Electrical Power Sources Unless otherwise specified herein, the equipment shall comply with Type II electrical power sources and connections requirements of MIL-T-28800D. The equipment shall operate from a nominal 115, single phase, 50/60 Hz power source.

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:58Z
Check the source to verify that this is the current version before use.

5 1 Fuses and Circuit Breakers Fuses and circuit breakers shall be in accordance with MIL-T-28800D A label clearly identifying fuse rating shall be located near the fuse holders Fuses shall be 1 25 inches (31 75mm) in length and 0 25 inches (6 35mm) in diameter

5 2 Grounding Grounding shall be in accordance with MIL-STD-1887, para 3 2 8

6. Accessories The following accessories shall be supplied with each equipment:

a One Government accepted operator's manual
b Input supply cable.
c Two insulated ground cables
d Shielded high voltage output cable.
f All other cables or adaptors necessary for initial turn-on or preoperational check, if required
g Other standard accessories, if any, that are offered commercially
h. A bag or pouch to stow all the accessories, if they cannot be stowed in the equipment front cover

7 Environmental Requirements Unless otherwise specified herein, the equipment shall comply with the Class 5 environmental requirements of MIL-T-28800D

7 1 Vibration The equipment shall comply with Class 5 vibration requirements of MIL-T-28800D, except that the equipment is not required to be operating during the vibration test

8 RAM Requirements.

8 1 Reliability requirements The equipment shall comply with the Type II Reliability requirements of MIL-T-28800D (paras 3 13 2 and 4 5 8 1 2) The contractor shall demonstrate his equipment MTBF (O) is equal to or greater than 1,500 hours Reliability tests shall be performed on the first production lot only

8.2 Calibration. Calibration interval shall meet or exceed 360 days with 85% of items still within tolerance at the end of the period.

8.3 Maintainability Requirements The contractor shall demonstrate his equipment Mean-Time-To-Repair (MTTR) (Ho in MIL-STD-471), is less than or equal to 90 minutes. The maximum tolerable MTTR (Hl in MIL-STD-471) shall be 115 minutes. The MTTR shall include all the time required to troubleshoot, fault isolate, repair and test the equipment for any malfunction down to the lowest discrete component (resistor, switch, transistor, integrated circuit, nonrepairable assembly, and so forth) of the equipment, but does not include calibration time

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T09:58Z
Check the source to verify that this is the current version before use.

9 Performance The following specifications apply to the equipment after a one hour warm-up

9.1 Output Voltage The output voltage shall be as a minimum continuously variable from 500 volts to 60,000 volts DC in no less than 3 ranges.

9 1 1 Indication The output voltage indication shall be by a separate 3 digit digital or analog meter that is fully protected against damage caused by test sample breakdown

9 1 1 1 Accuracy The accuracy of the indication shall be ±(2% of the reading + 100V) for digital indication or ±2% of full scale for analog indication

9.2 Output Current The output current shall be zero to five milliamps in no less than 4 ranges The maximum rated output shall be, as a minimum, 5 milliamps for 15 minutes

9 2 1 Indication The output current indication shall be by a separate 3 digit digital or analog meter that is fully protected against damage caused by test sample breakdown

9.2 1 1 Accuracy The accuracy of current indication shall be ±(2% of reading + 1 count) for digital indication or ±2% of full scale for analog indication

9 3 Ripple The peak to peak ripple in the output voltage shall be less than 2 5% at the maximum rated output value of five milliamps

9 4 Regulation The output voltage regulation shall be less than 20% from no-load to rated maximum output

9.6 Polarity The equipment shall be capable of providing either negative or positive polarity with reference to ground at the output terminal

9 7 Leakage Current The internal leakage current shall be less than 0 1 microamps at maximum rated output

9.8 Other Capabilities. The other performance capabilities of the equipment that are standard features of the commercial equipment, however, not covered by the above paragraphs of this CID.

10. Packaging Requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:58Z
Check the source to verify that this is the current version before use.

B QUALITY ASSURANCE PROVISIONS

1. Responsibility for inspection Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

2 Preconditioning Requirement (Burn-in) Each deliverable equipment shall be subjected to one of the following preconditioning requirements

a A minimum 48 hours on-time burn-in period prior to Acceptance Testing During the last 24 hours of on-time burn-in, the equipment shall operate failure free. Up until this time, the equipment will be allowed to accumulate failures Each equipment which fails during the final 24-hour period shall be repaired and returned to test until it successfully survives a 24-hour period without failure Failures which occur during the burn-in test shall be noted and reported, but shall not count toward the establishment of equipment MTBF Prior to burn-in the Satisfactory Operation Test (SOT) shall be conducted Daily Satisfactory Operation Checks (SOC) shall be conducted For the last 24-hour failure-free period, a complete SOT shall be conducted prior to and after the period The SOC and SOT are defined by MIL-T-28800D, paras 6 3 2 24 and 6.3 2.25 While under burn-in the equipment shall be subjected to the following conditions·

Ambient Temperature 40°C +5°C -0°C

Input Voltage Nominal specified voltage +5% -2%

Equipment On-Off Cycling. Equipment on for 8 hours and off for 4 hours. This cycle shall continue until a minimum 48 hours on-time is accumulated, with the last 24 hours of on-time, failure free.

Source: https://assist.dla.mil -- Downloaded: 202640-26T09:58Z
Check the source to verify that this is the current version before use.

b. Stress Screen. A minmum 15 minute vibration period followed by a minimum 10 thermal cycle period, shall constitute stress screen, prior to acceptance testing For the vibration period and the last three thermal cycles of stress screen, each equipment must survive failure free. Each equipment which fails during the vibration period or the last three thermal cycles shall be repaired and returned to test until it successfully survives these conditions. Failures which occur during the combined vibration/thermal stress screen shall be noted and reported but shall not count toward the establishment of equipment MTBF. An SOT shall be conducted prior to vibration and thermal cycling An SOT shall also be conducted during a 15 minute ambient temperature (25°C) test period, following each of the last three thermal cycles While under stress screen, the equipment shall be subjected to the following conditions.

Number of Thermal Cycles 10

Temperature Cycle· -40°C to +75°C

Temperature Rate of Change: 5°C/Min.

Input Voltage Nominal Specified Voltage +5%, -2%

Equipment On-Off Cycling. Equipment on at 0°C and off at 50°C Equipment shall not operate during chamber cool down

Vibration, Sinusoidal, linear sweep Sweep times shall be for a low to high to low frequency variation for each increment Equipment shall not operate during vibration

Frequency Increment Hz (High-Low)	Vibration Level (Inch Pk-Pk)	Vibration Time per Increment (Min)
5-15	0 06	7
15-25	0.04	3
25-55	0.02	5

Vibration Axis. Single axis, perpendicular to the plane of normal bench usage

3 Quality Conformance Inspection Unless otherwise required herein the test methods, procedures, and conditions used for quality conformance inspection shall be as as specified in MIL-T-28800D. Equipment shall be subjected to preconditioning requirement, para 2 above, prior to production inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

4. Production Inspection Each equipment shall be subjected to the Group A inspection of Table 1 to ensure qualitatively the proper functioning of the equipment, including all operating controls and conformance with safety requirements Group A testing (Table 1) is a reduced amount of testing Its purpose is to ensure that all functions and modes of operation of the equipment are evaluated without extensively checking each parameter as required in the Group B test (Table II)

4.1 Production Control Tests. The Group B tests of Table II shall be performed on equipments that have passed Group A inspection, para 4 above

4 1.1 Production Control Sampling Plan From each lot a random sample of equipment shall be selected and subjected to the Group B tests Sampling shall be in accordance with MIL-STD-105D, special inspection level S-3, AQL 1 5%

4.2 Environmental Tests The Group C tests of Table III shall be performed on equipments which have passed the required tests in accordance with para 4 1 above

4 2 1 Environmental Sampling Plan From the first production lot only a random sample of equipment shall be selected and subjected to the Group C tests The sampling shall be in accordance with MIL-STD-105D, special inspection S-1, AQL 4 0%

4 3 Reliability Tests The Group D tests of Table IV shall be performed on equipments which have passed the required tests in accordance with para 4 1 above

4.3.1 Reliability Tests Sampling Plan From the first production lot only a random sample of 10 equipments shall be selected and subjected to Group D tests

4 4 Other Tests. The Group E design tests of Table V shall be performed on equipments which have passed the required tests in accordance with para 4 1 above

4 4.1 Group E Design Tests Sampling Plan From the first production lot only, 1 equipment shall be randomly selected and subjected to the Group E tests with the exception of maintainability testing which requires a dedicated second sample

5. Reliability. The reliability tests shall be performed as follows:

5.1 Test Level. While under test, the equipment shall be subjected to conditions specified in para 4.5.8.1.2 of MIL-T-28800D (Test Plan IIID of MIL-STD-781D is invoked)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

5.2 Quality Conformance Inspection. The Group D reliability test shall be performed on the first production lot only From those units of the first production lot that have passed the required Group A and B tests, a random sample of ten (10) units shall be selected for Group D testing.

5.3 Test Length Testing shall continue until the total unit hours together with the total count of relevent equipment failures permit either an accept or reject decision Only equipment 'ON' time may be used in MTBF or longevity determinations Testing shall be monitored in such a manner that the time to failure may be estimated with reasonable accuracy No single equipment 'ON' time shall be less than one-half the average operating time of all equipments 'ON' time

5.4 Reliability Examination and Test Method The following inspections shall be used to verify equipment operation during reliability testing. The SOT of para 2 above shall be used to confirm proper equipment operation prior to and following reliability testing The SOT shall also be performed weekly to verify equipment operation. The SOC of para 2 above shall be used to monitor proper operation of the equipment daily

6 Maintainability Maintainability testing shall be conducted in accordance with para 4 5 7.1 1 of MIL-T-28800D.

TABLE I

GROUP A PERFORMANCE TESTS

REQUIREMENT	PARAGRAPH (Section A)
Safety	1
Personnel Hazards	1.1
Edge Bounding Exposed	1.2
Weight Caution Label	1 4
Marking	3.2.1
Marking and Identification	4
Supplemental Identification Plate	4 1
Information Plate	4.2
Battery Warning label	4.3
Electrical Power Sources	5
Fuses and Circuit Breakers	5.1
Accessories	6
Output Voltage	9.1
Indication	9.1.1
Output Current	9.2
Indication	9.2.1
Polarity	9.6
Other Capabilities (Selected Features)	9.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

TABLE II

GROUP B PERFORMANCE TESTS

REQUIREMENT	PARAGRAPH (Section A)
Accuracy	9.1 1.1
Accuracy	9.2.1.1
Ripple	9.3
Regulation	9 4
Leakage Current	9 7
Other Capabilities (those not included in Group A)	9 8

TABLE III

GROUP C ENVIRONMENTAL TESTS

Environmental Tests — MIL-T-28800D and Paragraph 7, Section A, of this Description

TEST	REQUIREMENT
Temperature, Non-operating	MIL-T-28800D
Temperature, Operating	MIL-T-28800D
Humidity	MIL-T-28800D
Altitude, Non-Operating	MIL-T-28800D
Vibration	MIL-T-28800D, see para 7 1 for exception
Functional Shock	MIL-T-28800D
Bench Handling	MIL-T-28800D
*Fungus Resistance	MIL-T-28800D
*Salt Exposure, Structural Parts	MIL-T-28800D

* Verification testing not required, but certificate of conformance is required

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

TABLE IV

GROUP D RELIABILITY TESTS

TEST	PARAGRAPH (Section A)
Reliability	8 1*

* Also para 5, section B

TABLE V

GROUP E DESIGN TESTS

REQUIREMENT	PARAGRAPH (Section A)
Radioactive Materials	1.3
Safety Features	1.5
System Safety Engineering Inspection	1 6
Design and Construction	2
Mainframe Plug-in Concept	2 1
Controls	2.2
Parts, Materials and Processes	3
Restricted Material	3 1
Interchangeability	3.2
Grounding	5.2
Maintainability	8.3
	PARAGRAPH (MIL-T-28800D)
Accessible Potentials	3.9 3.1
Ventilation opening	3.9 3 3
Equipment Opening	3 9 3.4
Thermal Hazard Protection	3 9.3.5
Warning Markings	3.9 3.6
Guards and Shields	3 9.3.7
Toxic Material	3 9.3.9
Compatibility of Material	3.9.3 10 1

MILITARY INTERESTS

Custodian.
Army - CR

CIVIL AGENCY COORDINATION ACTIVITY

GSA-FSS

PREPARING ACTIVITY:
Army - CR

(Project 6625-A962)

Source: https://assist.dla.mil -- Downloaded: 202610-26T09:58Z
Check the source to verify that this is the current version before use.